

**Uncertainty, Innovation, and Migration: A Macro-Level Empirical Investigation of
Inventor Mobility Under Uncertainty**

MGMT898 MSc Research Project

by

Steffin Shibu

(Student Number 20445424)

MSc Management - Business Economics

Under the Supervision of

Dr Olena Ivus

Associate Professor



Smith School of Business

Queen's University

Kingston, ON

Canada

July 2025

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Abstract

This study examines how macro-level uncertainty shapes international inventor migration. Contrary to the standard theoretical and empirical expectations, the results show that uncertainty can act as a pull factor, and not a deterrent, in inventor mobility. Using a global panel of 122 origin and 122 destination countries and a gravity-type model estimated via Poisson Pseudo Maximum Likelihood (PPML), we find that one-standard deviation (SD) increase in relative uncertainty at the destination leads to an increase in 19.7-25.2% in inventor flows in the subsequent window. This finding is robust to the inclusion of innovation-specific and macroeconomic controls and holds across alternative estimation strategies. The results challenge Real-Options Theory and recent push-based models by showing that inventors may actively move toward – rather than away from – volatile but opportunity-rich environments. In doing so, the study contributes a behavioural reinterpretation to the literature in skilled migration and the geography of innovation.

Keywords: Inventor migration, Relative uncertainty, Gravity-type model, PPML, Three-way fixed effects, WUI, Real-Options Theory.

Introduction

The international migration of inventors plays a central role in shaping the global geography of innovation and cross-border diffusion of knowledge. Unlike general skilled migration, inventor mobility entails not only the relocation of human capital but also transfer of tacit knowledge, research practices, and technology-specific capabilities.

These flows can alter regional comparative advantage, deepen transnational knowledge networks, and influence long-term trajectories of scientific productivity. Understanding who migrates, and under what conditions, is therefore of central importance to policymakers, firms, and scholars concerned with the future of innovation systems.

A growing body of empirical research has established that mobile inventors tend to be more productive, more internationally collaborative, and more integrated into global scientific networks than their non-mobile counterparts. Prior studies have also identified a range of structural and institutional “pull” factors that attract inventor talent, such as strong intellectual property rights, high expenditure on research and development (R&D), and world-class research infrastructure.

However, the role of macro-level uncertainty – economic, political, or institutional – remains poorly understood in these literatures. Most existing studies implicitly assume that uncertainty operates as a deterrent to mobility, discouraging investment and prompting skilled workers to seek more stable environments. This framing aligns with classical Real-Options Theory, which predicts that under uncertainty, rational agents delay action in favour of waiting for greater informational clarity (Bloom, 2009).

This theoretical framework may be too restrictive when applied to high-skilled, globally-mobile inventors. Unlike firms or capital, inventors possess internationally portable human capital and operate within distributed research ecosystems that span institutions, sectors, and national borders. Their mobility decisions often involve inter-temporal and

spatial arbitrage over opportunity landscapes. In this context, uncertainty may not uniformly deter movement. Instead, it may signal institutional reform, structural transformation, or even new funding opportunities – conditions that could render uncertain destinations more attractive, not less.

Recent empirical work has begun to explore this idea. Bisset et al. (2024), using patent-based mobility data from Europe, demonstrate that higher levels of economic policy uncertainty in origin countries are positively associated with the probability of international emigration among inventors. Their results are interpreted as evidence of a “push” mechanism, wherein inventors exit environments marked by instability.

This raises a deeper question: can uncertainty also operate as a pull factor? That is, will inventors move toward destinations that are more uncertain than their origin countries – especially if such uncertainty reflects dynamism, openness, or forward-looking investment in innovation? This study addresses these questions.

Using bilateral inventors migration flows between 122 origin and 122 destination countries, it evaluates how relative uncertainty, measured via the World Uncertainty Index – developed by Ahir et al. (2022) – shapes the direction and intensity of cross-border inventor movements.

Unlike previous studies that solely focus on the effects of origin-country risk, this analysis constructs a ratio-based measure of relative uncertainty to assess whether inventors systematically respond to volatility at the destination. In doing so, it reframes uncertainty not as a uniform barrier to mobility, but as a potentially attractive feature of dynamic innovation ecosystems.

To isolate the effect of uncertainty from other known drivers of migration, the model incorporates a set of lagged, bilateral controls. These include international collaboration, scholarly output, inventor stock, and number of citations. These variables reflect core

dimensions of research ecosystem and enable the estimation to net out the confounding influence of absorptive capacity, professional networks, and research opportunity. The identification strategy further leverages a high-dimensional fixed-effects structure to address time-invariant bilateral frictions and time-varying origin and destination shocks.

By centring uncertainty as an endogenous feature of migration decisions – rather than a background constraint – this study offers a new lens of the dynamics of global talent circulation. The findings complicate conventional wisdom that inventors flee instability, showing instead that uncertainty may enhance, rather than diminish, the attractiveness of certain destinations. In doing so, the analysis contributes to ongoing debates in migration theory, innovation policy, and institutional reform, particularly in an era marked by rising global volatility and intensifying competition for high-skilled talent.

Literature Review

The study of inventor migration has traditionally been situated within the broader literature on high-skilled mobility, yet it presents distinctive characteristics that warrant separate analytical treatment. Inventors are not merely knowledge workers; they are producers of frontier technologies, and their international movements have direct implications for the spatial distribution of innovation capacity and the cross-border diffusion of tacit knowledge.

Pioneering work by Miguelez and Fink (2013, 2017) introduced a global, patent-based methodology for tracking inventor migration through changes in PCT filing addresses. Their data revealed a sustained rise in cross-border inventor mobility since the mid-1980s, with flows that are far from random. Instead they follow patterned, path-dependent trajectories shaped by colonial ties, geographic proximity, and diasporic networks.

Akbaritabar et al. (2024) significantly expanded this approach by documenting bilateral inventor migration across 210 countries from 1998 to 2020 using Scopus data. Their

findings indicate that mobility is increasingly concentrated in innovation-intensive economies, and that migration patterns often mirror scientific collaboration networks and the international distribution of research output. The consequences of such mobility have been the subject of growing empirical attention.

Using an event-study design, Pellegrino et al. (2023) demonstrate that inventor migration is associated with substantial productivity gains. On average, inventors generate 0.23 more patent applications in the five years following relocation, with larger gains observed when moving to destinations with stronger intellectual property regimes and more developed research infrastructure.

Migueluez and Temgoua (2020) further show that knowledge flows are not unidirectional: origin countries also benefit, particularly through return migration, co-invention, and diasporic linkages. These findings support the view that inventor migration facilitates international knowledge recombination and technological spillovers, rather than simply reallocating talent from Global South to Global North.

However, scientific opportunity structures alone cannot fully explain inventor migration. Recent work emphasises the role of institutional frictions – particularly policy uncertainty and legal restrictions – in shaping mobility decisions. Soto Nishimura and Czaika (2024), in a meta-analysis of high-skilled migration studies, report consistent negative elasticities between migration policy restrictiveness and skilled inflows.

Similarly, Kahn and MacGarvie (2020), in their analysis of STEM PhDs trained in the United States, find that delays in permanent residency processing significantly reduce retention. For Chinese nationals, each additional year of delay reduces the probability of staying by 2.1 percentage points; for Indian nationals facing wait times exceeding five years, the likelihood of staying declines by over 10 percentage points. These results underscore the importance of institutional openness in retaining high-skilled migrants.

Despite these advances, the role of uncertainty – economic, political, or institutional – remains comparatively underexamined in the inventor migration literature. Classical models rooted in Real-Options Theory suggest that agents delay irreversible decisions under uncertainty to preserve future flexibility (Dixit & Pindyck, 1994; Bloom, 2009). Empirical evidence supports this logic: firms typically reduce or postpone R&D investment when faced with heightened economic policy uncertainty (Czarnitzki & Toole, 2011; Ivus & Wajda, 2018). Such models implicitly assume that uncertainty suppresses forward-looking action, functioning as a brake on both innovation and cross-border mobility.

However, this logic may not hold for individuals – particularly inventors – who possess internationally portable human capital and face relatively low sunk costs of relocation. Unlike firms, inventors operate across jurisdictions, collaborate within distributed global networks, and respond to a distinct set of constraints and incentives. Their behavioural responses to uncertainty may therefore diverge from firm-level decision-making.

Bisset et al. (2024) offer the first migration-specific test of this hypothesis. Using a binary choice model across twelve European countries, they find that a one-standard deviation (SD) increase in economic policy uncertainty at the origin raises the probability of inventor out-migration between 13.73% and 21.70%. Interpreted as push effect, these findings suggest that uncertainty can accelerate skilled emigration, particularly from institutional environments perceived as volatile or deteriorating.

Nevertheless, important limitations remain. The scope of analysis in Bisset et al. (2024) is confined to twelve European countries, where harmonised mobility regimes under the Schengen framework minimise institutional barriers to migration. Moreover, their study relies on Economic Policy Uncertainty (EPU) index, which captures policy-related volatility but omits broader macroeconomic and geopolitical risks addressed by the World Uncertainty

Index (WUI). Their empirical strategy also estimates the probability of leaving the origin country using inventor-level patent data, without modelling destination choice.

In addition, their approach defines relative uncertainty in comparison to a cross-sectional average, rather than a bilateral, dyadic condition. This framing emphasises push effects while leaving open the question of whether uncertainty at the destination also influences migration flows. This is particularly relevant in reforming or rapidly transforming economies, where uncertainty may signal potential upside rather than disorder. In such contexts, uncertainty may act as a pull factor, attracting risk-tolerant inventors seeking first-mover advantage in volatile but opportunity-rich environments.

This study addresses these gaps by extending the empirical frame to 122 origin and 122 destination countries, and by constructing a relative log uncertainty ratio using the WUI (Ahir et al., 2022). This ratio-based specification captures whether inventor flows increase when the destination is more uncertain than the origin – flipping the conventional logic of uncertainty as a deterrent. To isolate this effect, the analysis incorporates bilateral innovation-specific controls alongside a three-way fixed effects structure that accounts for unobserved country- and dyad-level shocks.

In doing so, the study contributes to a more nuanced understanding of how uncertainty interacts with the mobility of high-skilled human capital. It complicates the standard push-pull dichotomy by showing that, under certain conditions, uncertainty at destination may enhance – rather than hinder – the attractiveness of migration. This finding introduces a novel behavioural dimension to the migration-innovation nexus, suggesting that inventors, as globally mobile agents, may interpret volatility not only as risk, but also as signal and opportunity.

This reframing has implications not only for how migration decisions are theorised, but also for how destination countries position themselves in an increasingly competitive

global talent market. It highlights the importance of institutional signalling, where perceived openness to reform may attract innovation-seeking individuals even in the presence of volatility.

Methodology

Empirical Model

Gravity-type frameworks provide the work-horse structure for analysing bilateral migration flows of skilled workers. Following Beine et al. (2016) and Fink et al. (2017), the inventor movements are modelled as a function of relative push-pull forces between 122 origin and 122 destination countries, embedding the specification in the best-practice guidelines for high-dimensional gravity estimation set out by Yotov et al. (2016).

This study uses bilateral inventor migration flows between 1993 and 2012 based on PCT patent data constructed by Miguelez and Fink (2013, 2017), and aggregated over five-year windows. The functional form of the specification is:

$$Flow_{ijw} = \exp(\alpha_{ij} + \alpha_{iw} + \alpha_{jw} + \beta(rWUI_{ij(w-1)}) + \delta(rX_{ij(w-1)}) + \epsilon_{ijw})$$

where α_{ij} captures time-invariant bilateral factors (e.g., geographic distance, common language, colonial ties), α_{iw} represents origin-window fixed effects, and α_{jw} accounts for destination-window fixed effects. Together, these absorb all bilateral and country-specific heterogeneity relevant to migration flows.

The dependent variable $Flow_{ijw}$, denotes the cumulative inventor flow from origin country i to destination country j during five-year window w , and is defined as:

$$Flow_{ijw} = \sum_{t=n}^{n+4} Flow_{ijt}$$

where $n \in \{1993, 1998, 2003, 2008\}$ and $Flow_{ijt}$ counts the number of inventors from country i applying for patents in country j in year t .

The key explanatory variable is the relative log uncertainty ratio:

$$rWUI_{ij(w-1)} = \frac{\ln(WUI_{j(w-1)})}{\ln(WUI_{i(w-1)})}$$

constructed from the World Uncertainty Index and averaged over the preceding five-year window to mitigate the risk of simultaneity bias (Ahir et al., 2022; Fink et al., 2017). A positive coefficient (β) indicates that inventor flows increase when destination countries are relatively more uncertain than the origin countries, as measured by the ratio of their logged uncertainty indices.

All control variables – international collaboration, citations, number of authors, and scholarly output – are transformed into analogous relative log ratios, $rX_{ij(w-1)}$ to ensure the coefficients capture relative push-pull contrasts rather than absolute levels. This specification enables interpretation of coefficients in terms of directional effect of bilateral differences in innovation ecosystems.

Econometric Estimation

The model specified above is estimated using Poisson Pseudo Maximum Likelihood (PPML), the preferred estimator for gravity-type equations with heteroscedastic count data and structural zero flows (Head & Mayer, 2014). The inclusion of three-way fixed effects absorbs all time-invariant bilateral characteristics and time-varying origin- and destination-specific unobservable, such that identification arises from within-dyad variation in relative uncertainty and innovation conditions.

All explanatory variables are lagged by one window to mitigate simultaneity bias – uncertainty and research conditions measured in $(w - 1)$ can influence inventor movements realised in window w , but flows in w cannot feed back into already-lagged regressors. Standard errors are clustered at origin-destination pair level, allowing for arbitrary serial correlation within dyads.

As a robustness check, the specification is re-estimated using a panel regression with the Ordinary Least Squares (OLS) estimator. While OLS is less suited than PPML for zero-

inflated count data, it enables partial assessment of omitted variable bias, and concordant coefficient signs lend additional support to the PPML results. Further robustness is introduced by replacing discrete window values with moving-average smoothed regressors; these results are presented alongside baseline estimates in the results section.

To facilitate the interpretation, the coefficient on $rWUI$ is standardised such that a one-unit increase corresponds to a one-standard deviation (SD) change within the sample, following the approach in Bisset et al. (2024). Accordingly, β can be interpreted as the percentage change in inventor flows associated with a one-SD increase in relative log uncertainty, conditional on bilateral innovation differences and fixed effects.

Data and Variable Definitions

Inventor flows are sourced from bilateral migration matrix developed by Miguelez and Fink (2013, 2017), which records, for each origin-destination pair, the number of inventors whose nationality is country i and who subsequently relocate and apply for a patent in country j .

These bilateral counts are aggregated into four five-year windows (1993-1997, 1998-2002, 2003-2007, and 2008-2012), resulting in a balanced panel for 59,048 dyad-window observations covering 122 origin and 122 destination economies. An initial inspection of the raw data reveals several stylised facts that motivates the empirical strategy adopted in this study.

Table 1 presents the ten largest destination countries for migrant inventors across the four windows. The United States dominates consistently, absorbing over 85,000 inventors during the 2008-2012 period – more than eight times its intake during 1993-1997. Germany, Switzerland, and the United Kingdom also rank highly throughout, although their relative shares decline over time. Notably, China enters the top ten only in the final window, indicating an emergent role as a destination in the global innovation landscape.

Table 1 *Top Ten Destination Countries of Migrant Inventors during Four Five-Year Time Windows*

Country	Inflows	Country	Inflows	Country	Inflows	Country	Inflows
	1993-1997		1998-2002		2003-2007		2008-2012
US	10,346	US	36,354	US	97,829	US	85,579
Germany	2,636	Germany	6,364	Germany	11,713	Germany	11,377
UK	1,873	UK	4,972	Switzerland	9,360	Switzerland	10,714
Switzerland	1,833	Switzerland	4,792	UK	7,507	UK	7,096
France	1,211	France	2,591	Netherlands	4,772	France	4,188
Australia	867	Netherlands	2,369	France	4,617	China	4,177
Canada	739	Canada	1,955	Canada	3,471	Singapore	3,618
Belgium	674	Australia	1,605	Singapore	3,098	Netherlands	3,441
Sweden	510	Japan	1,452	Japan	2,726	Japan	2,930
Japan	435	Belgium	1,433	Belgium	2,337	Sweden	2,828

Table 2 presents the complementary perspective, listing the leading origin countries. While the United Kingdom and Germany dominate in the early periods, China and India exhibit significant growth, jointly accounting for a combined 48,000 outbound inventors by 2008-2012. These shifts reflect a broader reconfiguration of the global innovation system, with talent being increasingly sourced from emerging economies.

To explore how these national trends manifest at the bilateral level, Table 3 reports the ten busiest inventor migration corridors in the earliest and latest windows. The China-United States corridor expands more than tenfold – from just over 2,000 to nearly 23,000 migrants – emerging as the dominant global inventor mobility channel by the end of the sample period. Other major corridors, such as India-United States, Canada-United States, and Germany-Switzerland also grow, although their relative share of total flows narrows. These descriptive patterns underscore the need for a gravity-type framework that accounts for both persistent bilateral linkages and evolving global push-pull dynamics

Based on these empirical regularities observed in inventor flows, all explanatory and control variables are constructed using a relative log ratio specification. For each variable, the values are averaged over the preceding five-year window, denoted as $(w - 1)$, to ensure proper temporal ordering and to mitigate simultaneity bias. Subsequently, the values in the destination country are divided by the corresponding values in the origin country, both transformed by taking the natural logarithm.

The key regressor – relative log uncertainty ratio – is based on the World Uncertainty Index developed by Ahir et al. (2022). This index captures macroeconomic, political, and institutional uncertainty, as proxied by the frequency of uncertainty-related terms in Economist Intelligence Unit (EIU) country reports. Higher values of $rWUI$ imply that the destination country is more uncertain than the origin, allowing us to test whether inventors are drawn toward environments characterised by elevated macro-volatility.

Table 2 *Top Ten Origin Countries of Migrant Inventors during Four Five-Year Time Windows*

Country	Outflows	Country	Outflows	Country	Outflows	Country	Outflows
	1993-1997		1998-2002		2003-2007		2008-2012
UK	3,790	China	10,618	China	27,616	China	27,874
China	2,712	UK	8,281	India	21,602	India	20,470
Germany	2,549	Germany	8,180	Germany	16,691	Germany	14,816
India	1,823	India	6,536	UK	15,187	UK	11,607
US	1,258	France	4,416	Canada	11,325	Canada	10,543
Canada	1,116	Canada	4,229	France	9,087	France	10,122
France	1,057	US	3,012	US	5,397	US	5,912
Austria	883	Italy	2,307	Netherlands	4,966	Italy	5,604
Netherlands	831	Netherlands	2,159	Italy	4,517	Netherlands	3,856
Italy	722	Russia	2,026	Russia	4,247	Japan	3,310

Table 3 *Top Ten Bilateral Inventor Migration Corridors during Two Five-Year Time Window*

Origin	Destination	Flows	Origin	Destination	Flows
		1993-1997			2008-2012
China	US	2,039	China	US	22,885
UK	US	1,744	India	US	17,621
India	US	1,572	Canada	US	8,969
Canada	US	884	UK	US	6,334
Germany	Switzerland	697	Germany	US	4,624
Germany	US	691	Germany	Switzerland	4,177
Austria	Germany	619	France	US	3,102
UK	Germany	328	Japan	US	2,274
UK	Australia	289	France	Switzerland	1,872
Russia	US	280	Russia	US	1,573

In addition, several control variables are included to capture macroeconomic differences and structural contrasts in innovation ecosystems between destination and origin countries. These include international collaboration, total citations, scholarly output, number of authors, GDP per capita, and population. Data are drawn from SciVal, Scopus, Akbaritabar et al. (2024) and the World Bank database. Each control is constructed analogously to the key regressor: as relative log ratio, averaged across the same lagged five-year window.

Table 4 provides summary statistics for the key explanatory variables. The distribution of inventor flows is highly right skewed, with a mean of 7.03 and a maximum exceeding 23,000, justifying the use of PPML over OLS for estimation. Relative uncertainty exhibits a mean of 1.09 and a standard deviation (SD) of 0.48, indicating substantial asymmetries in macroeconomic stability across country pairs.

Table 4 *Descriptive Statistics*

Variable	Observations	Mean	SD	CoV*	Min	Max
Bilateral Inventor Migration Flow	59,048	7.03	203.93	29.02	0	23523
Relative Log Uncertainty	54,932	1.09	0.48	0.44	0.18	5.53
Relative Log Citation	44,286	1.12	0.81	0.73	0.04	23.20
Relative Log Int. Collaboration	44,165	1.21	1.16	0.95	-15.81	23.60
Relative Log Output	44,286	1.28	1.28	1	0.03	31.60
Relative Log Num of Inventors	43,566	1.21	0.87	0.72	0.07	15.27
Relative Log GDP Per Capita PPP	58,564	1	0.03	0.03	0.91	1.09
Relative Log Population	59,048	1.01	0.12	0.12	0.62	1.60

CoV* = Coefficient of Variation

Among the controls, relative international collaboration and scholarly output show the greatest dispersion, with coefficients of variation near or above 1. These patterns validate the inclusion of innovation-specific metrics in the gravity model to capture meaningful bilateral differences.

Finally, Table 5 presents the correlation matrix for the explanatory variables. As expected, the innovation-specific controls – including relative citations, collaboration, output, and inventor stock – exhibit strong pairwise correlations, frequently exceeding 0.90. Importantly, however, relative uncertainty is only weakly correlated with either of the control variables ($\rho \leq 0.16$), and is virtually uncorrelated with bilateral flows. This orthogonality suggests that the main coefficient of interest – β on $rWUI$ – is not confounded by underlying scientific fundamentals or unobserved institutional quality.

Following the standard econometric practice, multicollinearity among controls is acknowledged as a potential source of inflated standard errors; however, it does not bias the estimated effect of relative log uncertainty, which remains cleanly identified (Angrist & Pischke, 2009).

Taken together, the structure of the data provide a robust empirical foundation for assessing how macro-level volatility influence the global allocation of inventive talent. The windowed aggregation smooths annual fluctuations, while the relative log ratio specification converts country-level indicators into directional measures of migration incentives.

This design allows the analysis to isolate marginal responses to asymmetries in institutional conditions across countries, rather than relying on absolute levels. The broad country coverage further enables cross-regional generalisability, moving beyond OECD-centric migration research. As such, the dataset and empirical design jointly enhance both the credibility and external validity of the findings.

Table 5 *Correlation Matrix*

	1	2	3	4	5	6	7	8
1	1.000							
2	-0.009	1.000						
3	0.004	-0.129	1.000					
4	-0.003	-0.116	0.926	1.000				
5	-0.003	-0.100	0.904	0.911	1.000			
6	-0.003	-0.111	0.900	0.918	0.990	1.000		
7	0.007	0.007	-0.112	-0.104	-0.100	-0.105	1.000	
8	-0.001	-0.160	0.405	0.331	0.394	0.412	-0.094	1.000

Note: 1 = bilateral inventor flows; 2 = relative log uncertainty; 3 = relative log citations; 4 = relative log collaborations; 5 = relative log output; 6 = relative log num of inventors; 7 = relative log gdp per capita ppp; 8 = relative log population

Results

Relative Uncertainty as a Determinant

The central theoretical proposition of this study is that inventor migration responds systematically to cross-country differences in macro-level uncertainty. Specifically, the analysis tests whether inventors are more likely to migrate toward destinations that are relatively more uncertain than their countries of origin. This stands in contrast to conventional push-based models of migration under risk, and instead explores the possibility that uncertainty may act as a pull factor under certain structural or institutional conditions.

This proposition finds intuitive support in the broader literature on mobility under uncertainty, which emphasises that location choices are shaped not only by the absolute level of opportunity but also by the relative volatility of expected returns (Bloom, 2009; Bisset et al., 2024). In line with this reasoning, the analysis begins by testing whether higher relative uncertainty in the destination country, compared to origin, is associated with greater inventor flows.

Uncertainty is operationalised using the relative log ratio of the WUI between destination and origin countries, lagged by one window. The baseline model estimates its effect on bilateral inventor flows using a Poisson Pseudo Maximum Likelihood (PPML) framework.

Table 6 presents a stepwise estimation strategy, gradually introducing fixed effects to control for unobserved heterogeneity. Column (1) reports a naïve bivariate regression with no fixed effects. The coefficient on $rWUI$, is negative and weakly significant, suggesting a deterrent effect of uncertainty. However, this estimate is likely biased due to omitted variables correlated with both uncertainty and migration flows, such as crisis episodes or policy shocks.

Table 6 *PPML Estimation with Introduction of Fixed Effects*

	(1)	(2)	(3)	(4)
VARIABLES	No FE	Single FE	Two-way FE	Three-way FE
Relative Log Uncertainty	-0.531*	-0.795***	0.128	0.410*
	(0.278)	(0.286)	(0.110)	(0.210)
Constant	2.572***	7.638***	6.902***	6.636***
	(0.442)	(0.282)	(0.108)	(0.208)
Observations	54,932	8,873	8,738	8,517
Origin-Destination FE	No	Yes	Yes	Yes
Origin-Window FE	No	No	Yes	Yes
Destination-Window FE	No	No	No	Yes
Pseudo-R ²	0.00509	0.928	0.982	0.986

Robust standard errors in parentheses
 *** p<0.01, ** p<0.05, * p<0.1

In column (2), the inclusion of the origin-destination fixed effects absorbs time invariant bilateral characteristics – such as geographic distance, shared language, or colonial ties. The coefficient becomes more negative and statistically significant at the 1 percent level, reinforcing concerns about omitted variable bias in simpler specifications.

When origin-window fixed effects are added in Column (3), the coefficient on $rWUI$ becomes small and statistically insignificant. This highlights the importance of controlling for time-varying origin-country shocks – such as recessions or political disruptions – that may confound both inventor flows and uncertainty levels.

The key result emerges in Column (4), which includes the full set of three-way fixed effects: origin-destination, origin-window, and destination-window. This specification accounts for both time-invariant dyadic characteristics and time-varying country-specific

shocks. In this benchmark model, the coefficient on relative log uncertainty becomes positive, statistically significant at the 10 percent level, and economically meaningful.

The estimated coefficient of 0.410 implies that one-unit increase in the relative log ratio of destination-to-origin uncertainty in the preceding window is associated with a 41% increase in bilateral inventor flows in the current window.

To aid interpretation, the coefficient is scaled by the within-sample standard deviation of $rWUI$ ($SD = 0.48$), following Bisset et al. (2024). This yields an elasticity of approximately 19.7%, suggesting that a one-standard-deviation increase in relative uncertainty at the destination leads to a nearly 20% increase in inventor migration. This finding indicates that mobile inventors do not uniformly avoid uncertainty; under certain conditions, they may actively select into more volatile environments.

This result is corroborated and strengthened in Table 7, which introduces a sequence of innovation-specific controls while retaining the benchmark three-way fixed effects. Across all specifications, the coefficient on relative uncertainty remains positive, statistically significant, and remarkably stable – ranging from 0.491 to 0.526. These estimates imply that a one-standard-deviation (SD) increase in $rWUI$ is associated with a 23.5 to 25.2% increase in inventor flows.

Column (2) through (5) sequentially introduce controls for scientific collaboration, research ecosystem quality, innovation capacity, and potential migrant stock – each constructed as a relative log ratio. While multicollinearity limits the interpretability of individual control coefficients when entered jointly, Column (6) includes all controls simultaneously to isolate the net effect of relative uncertainty. The coefficient on $rWUI$ remains stable at 0.532, corresponding to a 25.5% increase in migration flows per SD . That this estimate does not attenuate in the full specification further supports the interpretation of uncertainty as a pull factor.

Table 7 *PPML Estimation with Innovation-specific Controls*

VARIABLES	(1) No Controls	(2) Network Effect	(3) Ecosystem Strength	(4) Potential Movers	(5) Ecosystem Quality	(6) All Controls
Relative Log Uncertainty	0.410* (0.210)	0.491** (0.225)	0.507** (0.225)	0.506** (0.225)	0.526** (0.223)	0.532** (0.222)
Relative Log Int. Collaboration		0.354 (0.362)				-1.025 (0.965)
Relative Log Output			1.098** (0.534)			1.774 (1.318)
Relative Log Num of Inventors				1.016 (0.658)		-0.648 (1.235)
Relative Log Citation					2.484 (1.819)	2.597 (3.402)
Constant	6.636*** (0.208)	6.265*** (0.465)	5.396*** (0.637)	5.508*** (0.767)	3.816* (2.072)	3.570 (3.141)
Observations	8,517	6,576	6,578	6,564	6,578	6,564
Three-way FE	Yes	Yes	Yes	Yes	Yes	Yes
Pseudo-R ²	0.986	0.987	0.987	0.987	0.987	0.987

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Taken together, the results from Table 6 and 7 indicate that inventor migration is not solely driven by risk aversion or a preference for stable environments. Instead, inventor flows respond meaningfully to uncertainty at the destination, suggesting that volatility may be interpreted as a signal of institutional reform, technological opportunity, or first-mover advantage. These findings challenge the standard Real-Options Theory framework and underscore the need to incorporate risk-tolerant behaviour into models of skilled migration under uncertainty.

Scientific and Economic Controls

To ensure that the estimated effect of relative uncertainty is not confounded by omitted drivers of inventor migration, the specification incorporates a series of innovation-specific and macroeconomic controls identified in prior literature. These include counts of international collaboration, scholarly output, citations, inventor stock, GDP per capita, and population.

The objective here is not to develop new theoretical claims regarding these covariates, but rather to demonstrate that the effect of relative uncertainty remains robust to their inclusion, and that the signs of their associations with bilateral inventor flows align with existing empirical findings. This approach functions as both a robustness check and a validation exercise for the empirical model.

As shown in Table 7, each innovation ecosystem variable is introduced sequentially while maintaining the full set of three-way fixed-effects. The stepwise inclusion mitigates omitted variable bias by conditioning on time-varying unobservable at both the origin and destination country levels. In every specification, the coefficient on relative log uncertainty remains positive, statistically significant, and stable in magnitude.

Individually, the innovation-related controls perform as expected: greater international collaboration, higher research output, a larger inventor pool, and more citations at the

destination (relative to origin) in the preceding window are all positively associated with inventor flows. These patterns are consistent with the findings of Fink et al. (2017) and Gorin (2016), who show that scientific connectivity, ecosystem strength, and technological reputation are significant predictors of inventor mobility.

Column (6) of Table 7 introduces all innovation-related controls simultaneously. While multicollinearity among these variables prevents clean interpretation of their individual coefficients, the stability of the $rWUI$ coefficient in this saturated model confirms that its effect is orthogonal to these ecosystem characteristics – highlighting its robustness to maximum conditioning.

Table 8 turns to the role of macroeconomic scale variables – GDP per capital and population – estimated under both PPML and OLS frameworks. In the PPML specification with three-way fixed effects, these variables are dropped due to near-perfect collinearity with country-window fixed effects. This is a known feature of high-dimensional gravity models and does not imply that macroeconomic scale is unimportant, only that its variation is absorbed by the fixed effects structure.

To investigate further, a parallel OLS model is estimated. Unlike PPML, OLS does not automatically exclude collinear variables, although it is less appropriate for heteroscedastic count data with a larger number of zero flows. In the OLS results, both GDP per capita and population enter with implausibly large and highly significant coefficients (1,967 and 5,689, respectively). These magnitudes likely reflect overfitting, multicollinearity, and scale distortion – issues also noted by Soto Nishimura and Czaika (2024) in the context of high-dimensional migration models.

Importantly, in the OLS specification, the coefficient on relative uncertainty remains positive and substantially larger (1.480), though statistically insignificant due to inflated standard errors. The directional consistency across estimators reinforces the conclusion that

Table 8 *PPML and OLS Estimation Including Macroeconomic Controls*

	(1)	(2)
VARIABLES	PPML	OLS
Relative Log Uncertainty	0.532** (0.222)	1.480 (3.061)
Relative Log Int. Collaboration	-1.025 (0.965)	-43.83*** (12.13)
Relative Log Output	1.774 (1.318)	30.97** (14.92)
Relative Log Num of Inventors	-0.648 (1.235)	-13.03 (14.37)
Relative Log Citation	2.597 (3.402)	215.4*** (58.13)
Relative Log GDP Per Capita PPP		1,967*** (708.0)
Relative Log Population		5,869*** (2,080)
Constant	3.570 (3.141)	-8,072*** (2,643)
Observations	6,564	43,080
Three-way FE	Yes	Yes
Pseudo-R ²	0.987	
Adjusted R ²		0.811

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

the $rWUI$ effect is not an artefact of estimator choice or of the exclusion of scale related variables.

Overall, the inclusion of scientific and macroeconomic controls confirms that the effect of destination-side uncertainty is robust, interpretable, and distinct from other known drivers of skilled migration. The signs and statistical significance of the control variables in the PPML specifications are broadly consistent with prior findings, lending external validity to the model design and further credibility to the interpretation of relative uncertainty as meaningful pull factor in global inventor mobility (Fink et al., 2017; Gorin, 2016).

Sensitivity and Robustness Checks

To assess the robustness of the estimated relationship between relative log uncertainty and inventor migration, the analysis implements two additional exercises: comparing results across estimation frameworks (Table 8) and evaluating the responsiveness of inventor flows to a smoothed, moving-average measure of uncertainty (Table 9).

The first robustness exercise compares PPML and OLS estimates in Table 8. In both models, the coefficient on relative log uncertainty remains positive. However, the OLS specification produces inflated standard errors and implausibly large coefficients on several covariates – driven by scale effects, multicollinearity, and overfitting. These results highlight the limitations of OLS in this context and reinforce the use of PPML as the preferred estimator. PPML accommodates heteroscedasticity, handles zero-flow dyads, and absorbs collinear country-window variation through high-dimensional fixed effects.

The second exercise, reported in Table 9, tests the temporal dimension of the inventor responses by replacing the lagged window average of relative uncertainty with a five-year moving average. This specification evaluates whether inventor migration responds to persistent macro-volatility or to more acute uncertainty shocks. Across all model variants, the coefficient on moving-average relative log uncertainty is small, statistically insignificant, and

Table 9 *PPML Estimation with Scientific Controls as Five-year Moving Averages*

VARIABLES	(1) No Controls	(2) Network Effect	(3) Ecosystem Strength	(4) Potential Movers	(5) Ecosystem Quality	(6) All Controls
Relative Log Uncertainty	0.0240 (0.196)	0.00756 (0.203)	0.0167 (0.202)	0.0269 (0.207)	0.0203 (0.199)	0.0227 (0.201)
Relative Log Int. Collaboration		0.230 (0.219)				-1.749 (2.160)
Relative Log Output			0.813 (0.537)			5.985 (3.808)
Relative Log Num of Inventors				0.633 (0.726)		-3.622 (2.362)
Relative Log Citation					1.293 (1.626)	-0.769 (4.758)
Relative Log GDP Per Capita						-
Relative Log Population						-
Constant	5.472* ** (0.189)	5.312*** (0.301)	4.641*** (0.615)	4.873*** (0.820)	4.106** (1.814)	5.735* (3.419)
Observations	39,029	32,452	32,458	30,573	32,458	30,567
Three-way FE	Yes	Yes	Yes	Yes	Yes	Yes
Pseudo-R ²	0.962	0.964	0.964	0.965	0.964	0.965

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

economically negligible. Even under the saturated specification, the point estimate implies a change of only 1.1% in expected inventor flows, based on a sample SD of 0.49.

These findings suggest that inventor migration is not driven by long-run institutional fragility or structural volatility. Rather, it is shaped by short-term, directional changes in relative uncertainty. In other words, inventors appear to respond to recent shocks in destination-country stability, rather than to background levels of chronic risk. This pattern is consistent with the temporal framing of Real Options Theory, which posits that agents are most responsive to short-run volatility. Such responsiveness points to recalibration of risk in high-skill labour markets, where volatility may be viewed less as a constraint and more as a temporal signal of future upside.

However, while real options logic predicts that volatility should induce delay in irreversible decisions, the results here indicate the opposite behavioural response: inventors treat short-run uncertainty as a signal of opportunity and exhibit greater mobility toward, not away from, volatile destinations. This divergence suggests that globally mobile inventors may not interpret uncertainty as a deterrent, but rather as a condition for strategic engagement. In doing so, inventors appear to leverage volatility as a positional advantage – seeking exposure to reform, entry-stage ecosystems, or shifting centres of global innovation.

Taken together, these sensitivity checks reinforce the credibility of the baseline findings. The estimated pull effect of relative uncertainty is not estimator-dependent, not confounded by scale variables, and not sensitive to temporal smoothing. Instead, inventor migration appears tightly linked to short-run uncertainty shocks at the destination. This pattern is robust across specifications and underscores the distinct behavioural logic that governs globally mobile innovation professionals.

These results challenge the canonical assumption of risk aversion embedded in traditional migration models. They support an alternative view of uncertainty as a forward-

looking attractor for innovative talent – a cue not of disorder, but of potential reform, dynamism, or first-mover advantage in high-skill global mobility.

Discussion and Conclusion

This study investigates the role of macroeconomic uncertainty as a determinant of international inventor migration. Contrary to prevailing assumptions that uncertainty deters mobility, results suggest that under certain institutional and behavioural conditions, uncertainty can act as a pull factor.

Specifically, the analysis finds that a one-SD increase in the relative log uncertainty ratio – defined as the ratio of lagged log WUI in the destination country to that in the origin – is associated with a 19.7% to 25.2% increase in bilateral inventor flows in the subsequent window. This effect is robust across multiple specifications, remains stable after controlling for innovation-specific and macroeconomic covariates, and is unaffected by estimator choice or temporal smoothing techniques.

At first glance, this finding appears to diverge from both Real-Options Theory and recent empirical work of Bisset et al. (2024). Real-Options Theory posit that, under conditions of uncertainty, economic agents delay irreversible investments in order to preserve flexibility (Dixit & Pindyck, 1994; Bloom, 2009). Similarly, Bisset et al. (2024) find that a higher economic policy uncertainty in the origin country significantly increases the probability of inventor emigration in Europe – an interpretation grounded in a push-based logic.

However, closer examination reveals that the present results are not inconsistent with these frameworks, but rather expand upon them. The two findings are theoretically coherent and empirically complementary.

First, the structural and geographic scope of this study differs substantially. While Bisset et al. (2024) focus on 12 Schengen countries, where cross-border mobility is facilitated by harmonised institutional regimes, this analysis covers 122 origin and 122 destination

economies – including many with restrictive visa systems, variable settlement barriers, and differing levels of institutional maturity.

In such heterogeneous contexts, uncertainty is less likely to be interpreted as disorder and more likely to be seen as a proxy for transformation, liberalisation, or market entry opportunity. For inventors with internationally portable human capital, these conditions may increase rather than decrease the appeal of moving.

Moreover, the finding that inventors respond positively to destination-side uncertainty introduces behavioural nuance often overlooked in canonical models. Rather than avoiding risk, mobile inventors may actively seek out dynamic environments where volatility signals institutional change or early-mover advantage. This perspective reframes uncertainty not as a deterrent but as a source of directional incentive – a possibility that becomes visible only when relative, dyadic comparisons replace absolute, origin-focused measures.

Second, the modelling strategy differs considerably. Bisset et al. (2024) employs a binary choice framework to estimate individual-level probabilities of exit from origin countries. In contrast, this study uses a high-dimensional gravity model to estimate bilateral inventor flows, capturing not only the decision to migrate but also the intensity of movement across dyads. As such, the dependent variable reflects aggregate talent flows, integrating both the propensity of individuals to migrate and the size of the inventor pool in each country.

While Bisset et al. (2024) interpret rising origin-side uncertainty as a push factor that increases the likelihood of inventor-level emigration, the macro-level evidence presented here shows that inventor flows also increase when the destination is relatively more uncertain. This suggests that inventors are not simply pushed from instability but may actively select into volatility – engaging with risk rather than avoiding it.

Third, the operationalisation of uncertainty also diverges. Bisset et al. (2024) rely on the Economic Policy Uncertainty (EPU) index, which reflects the media-based volatility in

fiscal and regulatory policy. This study instead uses World Uncertainty Index (WUI), which captures a broader spectrum of macroeconomic, institutional, and geopolitical uncertainty derived from EIU country reports (Ahir et al., 2022). The WUI thus better reflects systemic instability and potential reform – precisely the dimensions of uncertainty most relevant to high-skilled, career-oriented mobility.

Finally, while the findings challenge the behavioural predictions of Real-Options Theory – namely that agents delay action under uncertainty – they remain consistent with its temporal logic. The results show that inventor migration responds to lagged (short-term) shocks, but not to five-year moving averages of uncertainty. This suggests that inventors react to acute volatility rather than to persistent background fragility.

Crucially, however, they do not delay. Instead, they act swiftly in response to short-run shocks, treating uncertainty not as a signal to pause, but as an indicator of opportunity. In this sense, the findings reinterpret uncertainty as a forward-looking attractor in global talent markets – a signal of change, potential access, or reform.

Importantly, the estimated coefficient on relative uncertainty remains robust to the inclusion of standard innovation-specific and macroeconomic controls. Variables such as international collaboration, citation counts, inventor stock, and scholarly output behave as expected, consistent with prior findings (Fink et al., 2017; Gorin, 2016). The fact that the effect of relative uncertainty remains significant even in fully saturated specifications confirms that it is not driven by omitted fundamentals or correlated innovation dynamics, but instead reflects an independent behavioural response to shifting macro-risk landscapes.

These results open new theoretical avenues for understanding skilled-migration under uncertainty. They suggest that inventors – unlike firms or physical capital – possess structural and behavioural traits that enables them to navigate and potentially leverage volatility. Their internationally portable human capital, embeddedness in transnational research networks, and

low sunk costs of movement differentiate them from traditional economic agents embedded in Real-Options or Gravity-based frameworks. For such actors, uncertainty may signal first-mover advantage, nascent market formation, or reform-led opportunity structures.

However, while the empirical strategy is methodologically rigorous, the study is not without limitations. First, the use of three-way fixed effects, though essential for controlling unobserved heterogeneity, is inherently restrictive. It absorbs a substantial share of temporal and bilateral variation, limiting the scope for detecting interaction effects, dynamic adjustment processes or long-run spillovers.

Second, although the raw dataset comprises over 59,000 dyad-window observations, fewer than 9,000 are retained in the main PPML estimation due to data sparsity and missing values – particularly in earlier windows such as 1988-1992. Third, the high prevalence of zero flows, a common feature of migration data, further constrains the effective sample size, as fewer than 12% of observations are non-zero.

Future research could address these limitations through several avenues. First, more granular datasets such as Akbaritabar et al. (2024), which leverage Scopus-based author affiliations from 1998 to 2024, offer broader coverage of researcher migration beyond patenting activity. While these data target scientist rather than inventors, they represent a promising testbed for validating and extending the results.

Second, future work could explore behavioural heterogeneity by incorporating measures of individual-level characteristics such as risk tolerance, career stage, or the magnitude of migration-specific sunk costs. Third, further investigation is warranted into the conditions under which inventors derive first-mover advantage in volatile environments – potentially by linking migration trajectories to post-move patent productivity, grant success, or innovation outcomes.

In conclusion, the findings challenge dominant narratives of risk aversion in skilled migration. They demonstrate that uncertainty is not universally perceived as a deterrent; rather, it may act as a cue – a signal that change is imminent and opportunity is emerging. In an era of heightened global volatility, such insights are critical to understanding how innovative talent circulates, and how destination countries can position themselves as magnets for the world's most mobile and productive minds.

APPENDIX

List of Countries

OECD countries (31): Australia, Austria, Belgium, Canada, Chile, Colombia, Costa Rica, Denmark, Finland, France, Germany, Greece, Hungary, Ireland, Israel, Italy, Japan, Latvia, Lithuania, Mexico, Netherlands, New Zealand, Norway, Poland, Portugal, Slovenia, Spain, Sweden, Switzerland, United Kingdom, United States.

Non-OECD countries (91): Afghanistan, Albania, Algeria, Angola, Argentina, Armenia, Azerbaijan, Bangladesh, Benin, Bolivia, Bosnia and Herzegovina, Botswana, Brazil, Bulgaria, Burkina Faso, Burundi, Cambodia, Cameroon, Central African Republic, Chad, China, Croatia, Dominican Republic, Ecuador, El Salvador, Eritrea, Ethiopia, Gabon, Georgia, Ghana, Guatemala, Guinea, Guinea-Bissau, Honduras, India, Indonesia, Iraq, Jamaica, Jordan, Kazakhstan, Kenya, Kuwait, Lebanon, Lesotho, Liberia, Libya, Madagascar, Malawi, Malaysia, Mali, Mauritania, Moldova, Mongolia, Morocco, Mozambique, Myanmar, Namibia, Nepal, Nicaragua, Niger, Nigeria, Oman, Pakistan, Panama, Papua New Guinea, Paraguay, Peru, Philippines, Qatar, Russian Federation, Rwanda, Saudi Arabia, Senegal, Singapore, South Africa, Sri Lanka, Sudan, Tajikistan, Tanzania, Thailand, Togo, Tunisia, Turkmenistan, Uganda, Ukraine, United Arab Emirates, Uruguay, Uzbekistan, Viet Nam, Zambia, Zimbabwe

Notes

1. PCT stands for “Patent Cooperation Treaty”

$$2. Flow_{ijs}^{MA} = \frac{1}{5} * \sum_{s=t-4}^t Flow_{ijs}, \text{ where } t \in [1993, 2012]$$

3.

4.

5.

6.

7.

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